

Problem Set 5

1

The smart toilet's bidet has a micro-spray feature that sweeps a jet of water to help clean things up. The mechanism has sub-millimeter precision over the spray pattern owing to a micromachined Lorentz force actuator. The actuator is held in place by springs, and an equivalent model is shown for your analysis

a

Assume the springs are ideal and identical and support a 1 mm horizontal wire with a mass of 10 mg. The wire is energized with 100 mA of current (i_x). In the absence of a magnetic field the wire stretches the springs by 20 μm in the $-F_z$ direction due to gravity. When a maximum strength magnetic field is applied the springs stretch an additional 200 μm in the $-F_z$ direction. What is the value and direction of the magnetic flux density B ? Assume B is uniform across the wire and exerts no force on the springs.

Hint: the force of a spring is $F_z = k * d$ where the spring constant k is assumed to be constant and d is the stretch distance.

✓ Answer ✓

$$F_z = kd$$

$$d = 20 \mu\text{m}$$

$$g = 9.8 \text{ m/s}^2$$

$$m = 10 \text{ mg} = 10^{-5} \text{ kg}$$

$$F_g = mg = 9.8 \times 10^{-5} \text{ N}$$

$$F_g = F_z = kd$$

$$k = \frac{F_g}{d} = \frac{9.8}{2} = 4.9 \text{ N/m}$$

$$F_B = kd$$

$$d = 200 \mu\text{m}$$

$$F_B = 9.8 \times 10^{-4} \text{ N}$$

$$F_B = IL \times B \sin(\theta)$$

$$I = 100 \text{ mA} = 0.1 \text{ A}$$

$$L = 1 \text{ mm} = 0.001 \text{ m}$$

$$B = \frac{F_B}{IL} = \frac{9.8}{0.1(0.001)} \times 10^{-4} = 9.8 \text{ T}$$

Since F_B is in the $-\hat{z}$ direction, the current is in the $+\hat{x}$ direction, B should point in the $-\hat{y}$ direction as $F = L \times B$.

$$B = -9.8\hat{y} \text{ T}$$

A 20-turn rectangular coil with sides $l = 30$ cm and $w = 10$ cm is placed in the $y - z$ plane as shown in Fig. P5.6.

a

If the coil, which carries a current $I = 10$ A, is in the presence of a magnetic flux density

$$\vec{B} = 2 \times 10^{-2}(\hat{x} + \hat{y}2) \text{ (T)}.$$

Determine the torque acting on the coil.

✓ **Answer**

$$\tau = m \times B$$

$$m = NIA$$

$$N = 20$$

$$I = 10 \text{ A}$$

$$A = 3 \times 10^{-2}(\pm \hat{x}) \text{ m}^2$$

$$m = 6 \text{ Am}^2$$

$$\tau = \pm 0.06 \hat{z}$$

b

At what angle ϕ is the torque zero?

✓ **Answer**

$$B = a(\hat{x} + \hat{y}2)$$

$$m \times B = 0$$

$$m = b(\hat{x} + 2\hat{y})$$

$$\phi = \arctan(2, 1) = 1.1071487177940904$$

c

At what angle ϕ is the torque maximum? Determine its value.

✓ **Answer**

$$B = a(\hat{x} + \hat{y}2)$$

$$m = b(-2\hat{x} + \hat{y})$$

$$\phi = \arctan(1, -2) = 2.677945044588987$$

3

An infinitely long wire carrying a 25 A current in the positive x direction is placed along the x axis in the vicinity of a 20-turn circular loop located in the $x - y$ plane (Fig. P5.11). If the magnetic field at the center of the loop is zero, what is the direction and magnitude of the current flowing in the loop?

✓ Answer

$$\begin{aligned}
 B_{\text{wire}} &= \frac{\mu_0 I_{\text{wire}}}{2\pi d} \\
 &= \frac{\mu_0}{\pi} \frac{25}{4} \text{ T} \\
 B_{\text{loop}} &= \frac{N\mu_0 I_{\text{loop}}}{2R} \\
 &= \mu_0 10I \text{ T} \\
 I &= \frac{25}{40\pi} \text{ clockwise from the viewpoint}
 \end{aligned}$$

4

In a certain conducting region, the magnetic field is given in cylindrical coordinates by

$$\vec{H} = \hat{\phi} \frac{4}{r} [1 - (1 + 3r)e^{-3r}]$$

Find the current density $\vec{J}$.

✓ Answer

$$\vec{J} = \nabla \times \vec{H}$$

$$\frac{d\vec{H}}{dr} = 12e^{-3r} - 12e^{-3r} + 36re^{-3r} = 36re^{-3r}$$

$$\vec{J} = 36re^{-3r} \hat{z} \text{ A/m}^2$$

5

A thin current element extending between $z = -L/2$ and $z = L/2$ carries a current I along $+\hat{z}$ through a circular cross-section of radius a .

a

Find $\vec{A}$ at a point P located very far from the origin (assume R is so much larger than L that point P may be considered to be at approximately the same distance from every point along the current element).

✓ Answer

$$\begin{aligned}
 &\frac{\mu_0}{4\pi} \int \frac{I}{R} dl \\
 &= \frac{\mu_0 IL}{4\pi R} \hat{z}
 \end{aligned}$$

b

Determine the corresponding $\vec{H}$.

✓ Answer

$$\mathbf{H} = \nabla \times \frac{\mathbf{A}}{\mu_0}$$

$$A = \frac{\mu_0 I L}{4\pi R} (\hat{R} \cos \theta - \hat{\theta} \sin \theta)$$

$$\mathbf{H} = \hat{\phi} \frac{I L \sin \theta}{4\pi R^2} \text{ A/m}$$